Urban Informatics

Assignment - Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total marks: (10 X 2) = 20 marks**

MCQ/MSQ Question

QUESTION 1: [2 marks]
In IoT communication, ______ provides a unique address to devices, while ______ ensures reliable data transmission between devices and servers.

a. HTTP, MQTT
b. IP, TCP/IP
c. CoAP, WebSockets
d. MQTT, HTTP

Correct Answer: b. IP, TCP/IP

Detailed explanation: IP assigns a unique address; TCP/IP ensures reliable communication. (*Refer to Lecture 51 Slide 5*)

QUESTION 2: [2 marks]
Which component in a Computer Process Control System (CPCS) is responsible for converting analog signals into digital form for processing?

a. DAC
b. Controller
c. ADC
d. Actuator

Correct Answer: c. ADC

Detailed explanation: ADC converts analog signals into digital form. (*Refer to Lecture 52 Slide 3*)

QUESTION 3: [2 marks]
Match the IoT layer with its function:

IOT Layer	Function
P. Sensing Layer	i. Data transmission
Q. Network Layer	ii. Real-world data capture
R. WPAN	iii. Short-range communication
S. WWAN	iv. Long-distance communication

a. P-ii, Q-i, R-iii, S-iv
b. P-i, Q-ii, R-iv, S-iii
c. P-ii, Q-iii, R-i, S-iv
d. P-iii, Q-i, R-ii, S-iv

Correct Answer: a. P-ii, Q-i, R-iii, S-iv

Detailed explanation: Sensing layer captures real-world data, Network layer transmits data, WPAN is short-range (Bluetooth, Zigbee), WWAN is long-range (4G/5G). (*Refer to Lecture 52 Slide 4*)

QUESTION 4: [2 marks]
Which of the following is a primary purpose of the Arduino IDE?

a. To manufacture microcontrollers
b. To write, compile, and upload programs to microcontrollers
c. To design hardware circuits only
d. To replace embedded systems

Correct Answer: b. To write, compile, and upload programs to microcontrollers

Detailed explanation: Arduino IDE is used for coding and uploading programs. (*Refer to Lecture 52 Slide 9*)

QUESTION 5: [2 marks]
Which of the following statements correctly distinguishes analog and digital signals in terms of noise and representation?

a. Analog signals are discrete and noise-tolerant
b. Digital signals are continuous and noise-sensitive
c. Analog signals are continuous and noise-sensitive, while digital signals are discrete and more noise-tolerant
d. Digital signals have infinite values and analog signals have finite values

Correct Answer: c. Analog signals are continuous and noise-sensitive, while digital signals are discrete and more noise-tolerant

Detailed explanation: Analog is continuous, digital is discrete. (*Refer to Lecture 53 Slide 3*)

QUESTION 6: [2 marks]
In a sensor system, the transfer function is represented as ______, where 's' denotes ______.

a. S=f(s), stimulus
b. $S=s^2$, output signal
c. f=S(s), noise
d. S=m+c, sensitivity

Correct Answer: a. S=f(s), stimulus

Detailed explanation: The transfer function relates input stimulus (s) to output signal (S). It defines how a sensor converts physical input into electrical output. (*Refer to Lecture 53 Slide 6*)

QUESTION 7: [2 marks]
What is the primary function of an actuator?

a. Measure physical parameters
b. Store data
c. Convert electrical signals into physical action
d. Transmit data over network

Correct Answer: c. Convert electrical signals into physical action

Detailed explanation: Actuators take control signals and convert them into physical actions like motion, pressure, or sound. (*Refer to Lecture 53 Slide 11*)

QUESTION 8: [2 marks]
Which microcontroller is used as the main processing unit in Arduino Uno?

a. ATmega2560
b. ATmega328P
c. ESP32
d. PIC16F877A

Correct Answer: b. ATmega328P

Detailed explanation: Arduino Uno uses ATmega328P as microcontroller. (*Refer to Lecture 54 Slide 11*)

QUESTION 9: [2 marks]
Match the Arduino library with its function:

Arduino Library	Functions
P. EEPROM	i. Communication
Q. Ethernet	ii. Data storage
R. GSM	iii. Network connectivity
S. Bridge	iv. Processor communication

a. P-ii, Q-iii, R-i, S-iv
b. P-iii, Q-ii, R-iv, S-i
c. P-i, Q-ii, R-iii, S-iv
d. P-iv, Q-i, R-ii, S-iii

Correct Answer: a. P-ii, Q-iii, R-i, S-iv

Detailed explanation: EEPROM enables storage, Ethernet is used for network, GSM is used for communication, Bridge enables processor interaction. (*Refer to Lecture 55 Slide 7*)

QUESTION 10: [2 marks]
Which step is essential after installing Arduino IDE for proper board detection?

a. Writing code
b. Installing USB drivers
c. Uploading firmware
d. Connecting sensors

Correct Answer: b. Installing USB drivers

Detailed explanation: USB drivers are necessary so that the computer can detect and communicate with the Arduino board. (*Refer to Lecture 55 Slide 3*)
